PROFESSIONAL ENGINEERING PORTFOLIO

Caoilte Donohoe

Electronic Systems & Product Engineering

25+ years in hardware, embedded systems, RF and manufacturing

FIVE CORE ENGINEERING CAPABILITIES

01 **Electronic Hardware & PCB Design**

Analog, mixed-signal, RF and multilayer PCB development

02 **Embedded Firmware & Connected Systems**

ESP32, STM32, Nordic, cellular, BLE and IoT

03 **Energy Monitoring & Instrumentation**

Three-phase metering, signal conditioning and validation

04 **Verification & Product Integration**

Simulation, bench testing, troubleshooting and field pilots

05 **NPI & Volume Manufacturing**

DFM/DFT, production test, suppliers and change control

SELECTED COMMERCIAL ENGINEERING EXPERIENCE

FAAC Electronics | Watt Footprint | AmbiSense

25 pilot units tested | Designs reaching ~10,000 units/week

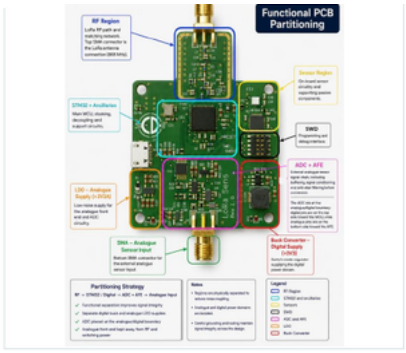
GITHUB ENGINEERING PORTFOLIO

github.com/CDonohoe-Designs/PCB-Electronics-Engineering-Portfolio

ENGINEERING CAPABILITY & COMMERCIAL OUTCOMES

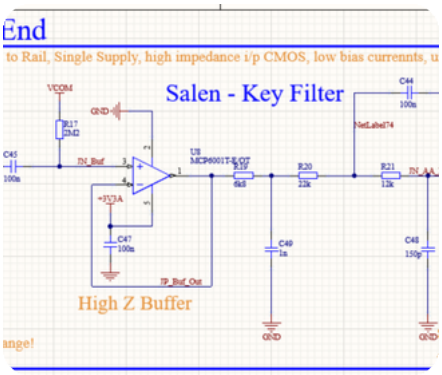
From engineering detail to product delivery

Hands-on technical depth combined with commercial NPI and manufacturing experience.



Functional partitioning of RF, MCU, sensing, ADC/AFE and separate

SYSTEM ARCHITECTURE



CIRCUIT IMPLEMENTATION



PRODUCTION HANDOFF

SELECTED COMMERCIAL RESULTS

~10,000 units/week

FAAC ELECTRONICS

RF/control designs supported through trial builds, pre-series production, test documentation and manufacturing support.

25 pilot units

WATT FOOTPRINT

12-channel energy monitoring with cellular connectivity, metering verification and customer-pilot deployment.

~5 to ~30 days

AMBISENSE

Battery-life improvement through power-cycle analysis, hardware changes and reliability investigations.

END-TO-END ENGINEERING RESPONSIBILITY

Requirements > Architecture > Design > Integration

Verification > NPI > Manufacturing > Field support

COMMERCIAL ENGINEERING / PRODUCT INTRODUCTION

RF Hardware, NPI & Volume Manufacturing

FAAC Electronics | R&D, design qualification and manufacturing support

~10,000
units/week for selected designs

50-100
unit trial builds

~1,000
unit pre-series builds

THREE DIMENSIONS OF ENGINEERING OWNERSHIP

01 PRODUCT DEVELOPMENT

RF TX / RX
433 / 868 MHz

v

CONTROL
Electronic design

v

DESIGN RELEASE
Production records

02 QUALIFICATION & NPI

BENCH & RF
VNA / spectrum tests

v

ENVIRONMENT
Temp / humidity / drop

v

PRODUCTION
Test specs / trials

03 FAILURE INVESTIGATION

OBSERVE
RF or PCB failure

v

TRACE CAUSE
Part / plating change

v

CORRECT
Supplier action

COMMERCIAL ENGINEERING CONTRIBUTION

Designed and revised RF transmitter, receiver and control electronics within an ISO 9001 manufacturing environment. Worked across R&D, purchasing, incoming inspection, quality, test, production and document control to support controlled introduction and product changes.

EXAMPLES OF ENGINEERING JUDGEMENT

Investigated RF performance degradation traced to an inferior substitute inductor. Separately identified dendritic-growth shorts following a PCB fabrication plating change, leading to a production halt and supplier corrective action.

From Concept to Customer Pilot

Watt Footprint | Hardware, embedded firmware and product integration

12 CHANNELS

4 three-phase circuits

25 UNITS

Built and tested

3 UNITS

Field tested

MULTI-CHANNEL METERING HARDWARE & FIRMWARE

SYSTEM SPECIFICATION

Metering: 12 single-phase / 4 three-phase

Metering ICs: 4 x ATM90E36

Controller: ESP32

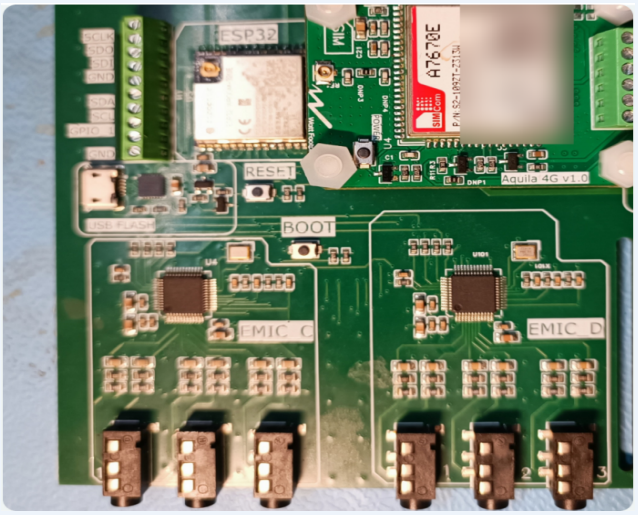
Connectivity: A7670E cellular; Wi-Fi / BLE

Telemetry: MQTT / JSON

Data: SD logging and replay

Installer configuration: BLE interface

Power: 5 V / 3.3 V, LiPo



Original hardware: metering and cellular integration

END-TO-END ENGINEERING OWNERSHIP

Progressed the product from initial concept through hardware design, embedded firmware development and system integration to a working customer-pilot build. Developed ESP32 firmware for four ATM90E36 ICs (12 channels), with cellular MQTT/JSON telemetry, watchdog supervision, modem recovery and remote management; integrated metering, local buffering and installer configuration.

PILOT BUILD, TESTING & FIELD VALIDATION

Built and tested 25 pilot units, including integrated hardware and firmware checks. Three of those units were field tested, providing real-world pilot exposure. Metering readings showed approximately +/-0.5% agreement with a calibrated Fluke reference under the reported test conditions.

PRODUCT DELIVERY OUTCOME

Concept -> electronic design + firmware -> integrated testing -> 25 pilot units tested -> 3 units field tested. Pilot hardware cost was approximately EUR150 including CT clamps, against a sub-EUR200 target.

ELECTRICAL MEASUREMENT PARAMETERS | ATM90E36

Per-phase RMS voltage/current; active, reactive and apparent power; energy; power factor; line frequency and phase-angle information. THD not included in this project.

Power, Reliability & Multi-Point Sensing

AmbiSense | Battery-powered cellular environmental and gas monitoring

~5 days
original operation

~30 days
revised operation

~6x
runtime increase

THREE CONNECTED ENGINEERING CONTRIBUTIONS

01 POWER OPTIMISATION

- Profile complete operating cycle: pump, modem and standby
- Change lead-acid battery to Li-ion
- Reduce pump duty; introduce latching relay
- Check gas-measurement performance

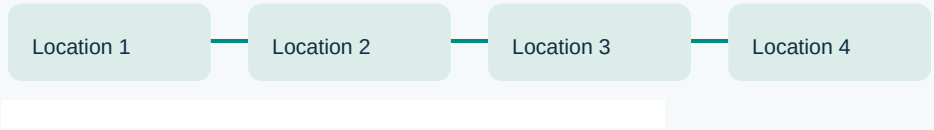
02 FOUR-POINT SAMPLING

- Design a four-channel multiplexed sampling arrangement
- Monitor gas at four different locations
- Share the existing gas-sensor monitoring hardware
- Select sampling locations sequentially

03 RELIABILITY & NPI

- Investigate boost-converter faults and modem resets
- Trace root cause to PCB layout
- Build, test and deploy replacement power hardware
- Support practical NPI and quality controls

FOUR-CHANNEL MULTIPLEXED GAS SAMPLING — CONCEPTUAL ARRANGEMENT



ENGINEERING CONTRIBUTION

Extended battery operation from approximately five to thirty days by measuring system loads and redesigning battery and switching arrangements. The increased autonomy supported alternative sensing approaches, including a four-channel multiplexed system that sampled four locations using shared gas-monitoring hardware.

Also diagnosed PCB-layout-related power faults and cellular-modem resets; replacement power hardware was built, tested and deployed, with no further related failures reported during the period of involvement.

ENGINEERING OUTCOME

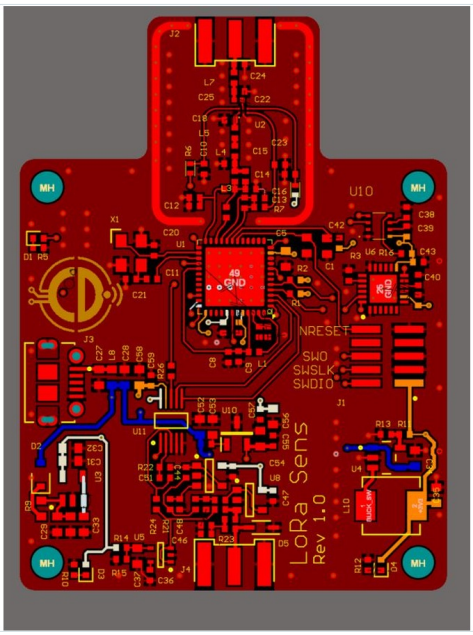
Improved operating autonomy, expanded multi-location sampling capability and resolved recurring power-system faults.

MIXED-SIGNAL AND RF HARDWARE

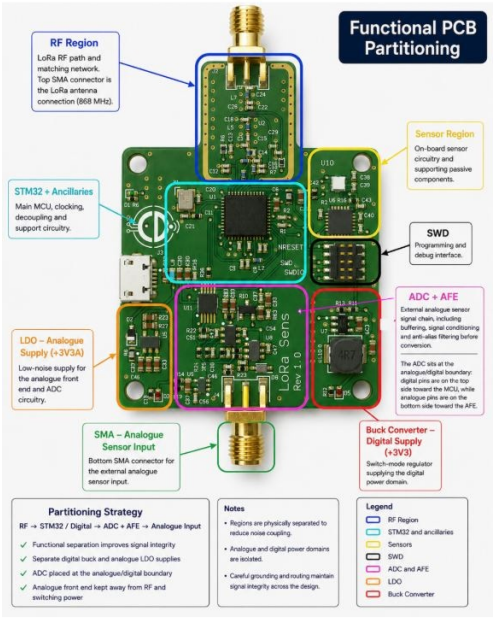
STM32WL · Mixed-Signal & RF Circuit Design

Analog signal conditioning, precision ADC acquisition and 868 MHz RF integration on a four-layer Altium PCB.

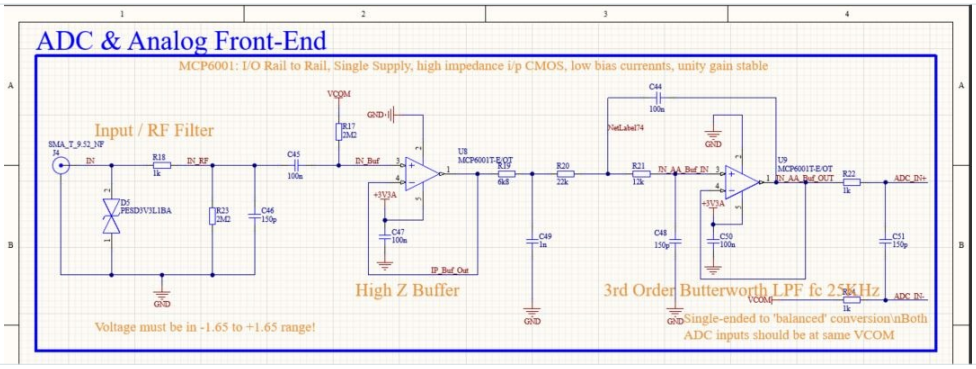
A four-layer sensor node combining an analog acquisition chain, precision conversion, separate



Altium copper layout: RF at the antenna end; MCU centrally; analog and power circuitry below.



Functional partitioning of RF, MCU, sensing, ADC/AFE and separate supply domains. Annotated board visualisation.



Protected SMA input, high-impedance buffer and active anti-alias filter from the Altium schematic.

Analog acquisition and reference

MCP6001 buffering and a third-order low-pass filter precede the ADC141S626 14-bit SPI ADC. REF3033 supplies the 3.3 V reference; VCOM provides the signal bias.

Power, RF and PCB implementation

TLV62569 digital supply and LD3985 analog LDO share a continuous ground reference. BGS12SN6 switching, matching/filtering, a dedicated RF net class and via fencing support the RF section.

Verification: schematic and layout completed; analog and RF simulations documented; Altium DRC snapshot reports 0 warnings and 0 violations.

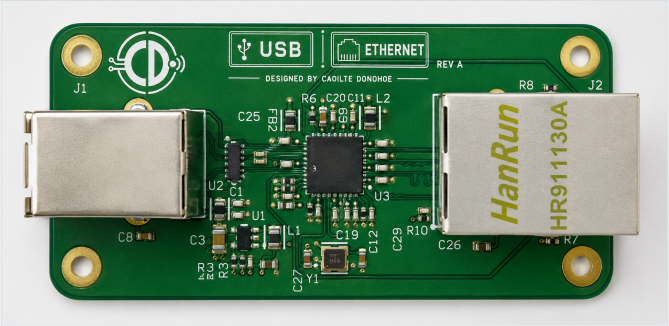
Detailed STM32WLE5 hardware report

Source: STM32WLE5 Hardware Design Report Rev 2.0, sections 2-8 and 11; CAD and rendered visuals.

HIGH-SPEED DIGITAL AND PCB ENGINEERING

USB 3.0 / Gigabit Ethernet · High-Speed PCB Design

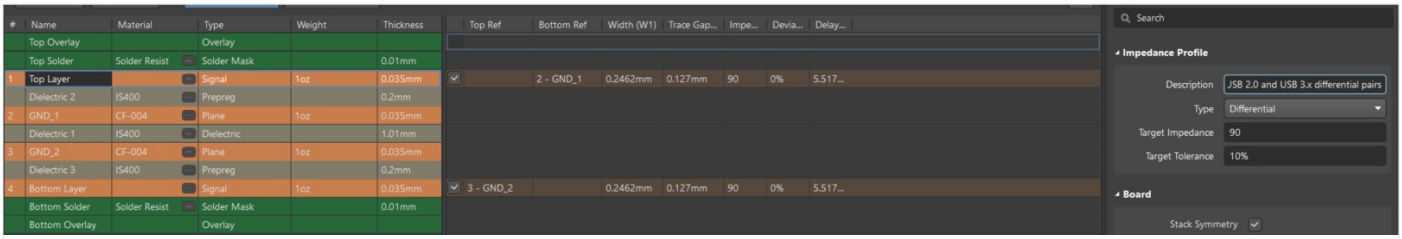
LAN7800 design using controlled impedance, differential-pair rules, layer-stack constraints and documented DRC.



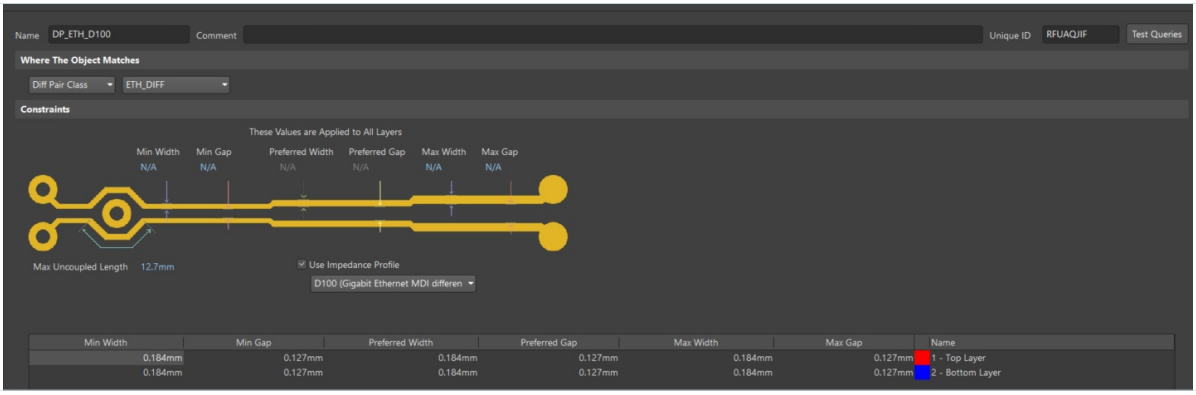
LAN7800 board visualisation from the existing project portfolio.

Constraint-driven layout

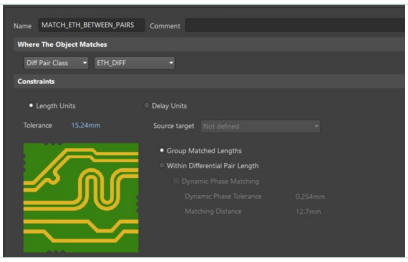
USB uses a 90 ohm differential profile. Ethernet has a separate D100 profile and ETH_DIFF routing scope. Named net classes separate USB2_HS, USB3_SS, ETH_MDI, clock, power and switch nodes.



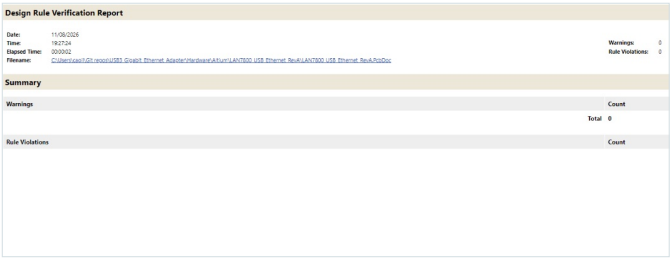
USB profile in Altium: 90 ohm target, 10% tolerance, 0.2462 mm trace width and 0.127 mm pair gap in this snapshot.



Ethernet differential-pair rule linked to its impedance profile; layer-specific width and gap values are explicit.



Matched-length rule scoped to ETH_DIFF. A separate rule controls within-pair matching.



USB/Ethernet DRC report snapshot: 0 warnings and 0 rule violations for the configured checks.

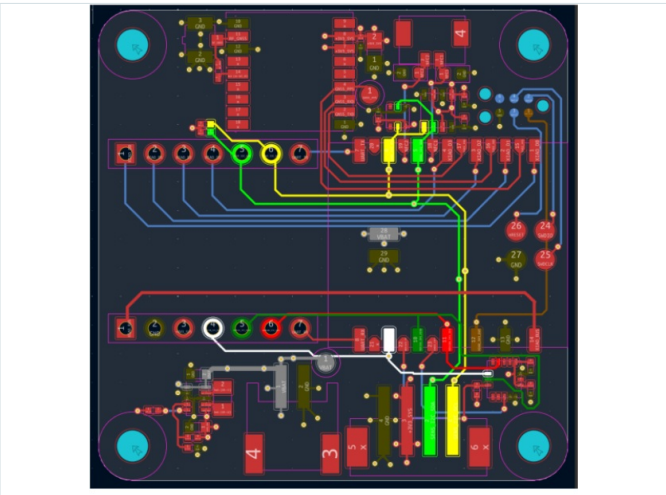
Engineering evidence: the layer-stack snapshot uses outer signal layers and two internal ground planes. Routing rules connect the electrical targets to PCB geometry. DRC results document the configured checks.

[USB3-Gigabit Ethernet design repository]

Sources: supplied USB Altium stack, impedance, net-class, differential and length-rule screenshots; DRC dated 11 Aug 2026.

EMBEDDED FIRMWARE AND BLUETOOTH LE

Nordic nRF52840 · BLE & Zephyr Integration



Custom Nordic PCB layout in KiCad: GNSS, power, sensor and expansion interfaces.

```
CCC_CALLBACK(humidity)
CCC_CALLBACK(pressure)
CCC_CALLBACK(gas)
CCC_CALLBACK(dew)
CCC_CALLBACK(record)
CCC_CALLBACK(light)

#define MEASUREMENT(uuid, index, callback) \
    BT_GATT_CHARACTERISTIC(uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY, \
        BT_GATT_PERM_READ, read_measurement, NULL, (void *)&measurement_indices[index]), \
    BT_GATT_CCC(callback, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)

BT_GATT_SERVICE_DEFINE(environmental_service,
    BT_GATT_PRIMARY_SERVICE(&service_uuid),
    MEASUREMENT(&temperature_uuid.uuid, 0, temperature_ccc),
    MEASUREMENT(&humidity_uuid.uuid, 1, humidity_ccc),
    MEASUREMENT(&pressure_uuid.uuid, 2, pressure_ccc),
    MEASUREMENT(&gas_uuid.uuid, 3, gas_ccc),
    MEASUREMENT(&dew_uuid.uuid, 4, dew_ccc),
    BT_GATT_CHARACTERISTIC(&record_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_record, NULL, NULL),
    BT_GATT_CCC(record_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE),
    BT_GATT_CHARACTERISTIC(&interval_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_WRITE,
        BT_GATT_PERM_READ | BT_GATT_PERM_WRITE, read_interval, write_interval, NULL),
    BT_GATT_CHARACTERISTIC(&light_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_light, NULL, NULL),
    BT_GATT_CCC(light_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)
);

static const struct bt_data advertising[] = {
    BT_DATA_BYTES(BT_DATA_FLAGS, BT_LE_AD_GENERAL | BT_LE_AD_NO_BREDR),
    BT_DATA_BYTES(BT_DATA_UUID128_ALL, ENV_UUID(0x8f3a0000)),
};
```

Zephyr custom GATT implementation: READ/NOTIFY, CCC subscriptions and writable configuration.

Hardware design

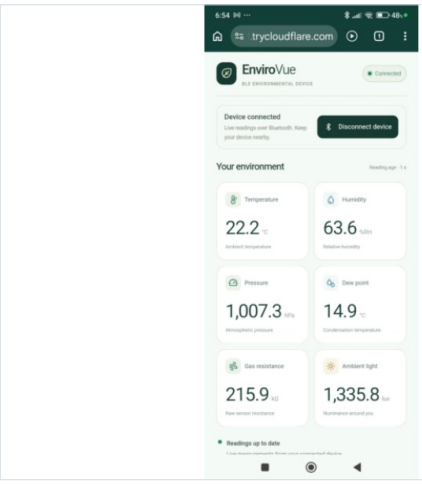
MAX-M10S GNSS and active-antenna interface; NFC connections; TPS63802 battery regulation; sensor integration, SWD and modular I2C/SPI/UART expansion.

Firmware implementation

Modular sensor, BLE, display and application code; Zephyr configuration and logging; persistent settings; custom 128-bit service and writable update interval.



Physical Nordic development setup with live OLED sensor values and BLE connection state.



EnviroVue client showing live measurements received over BLE.

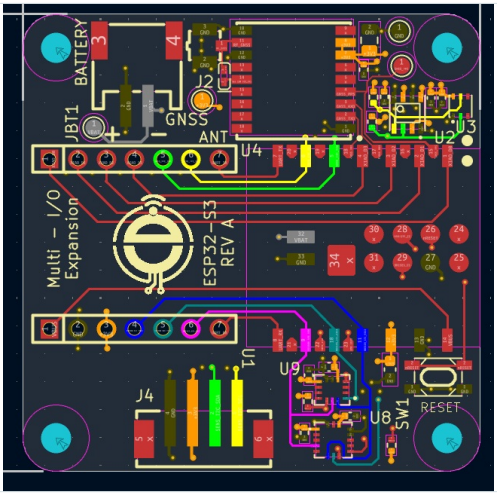
Evidence scope: custom PCB design and manufacturing preparation are separate from the physical development-board firmware demonstration.

[Nordic BLE engineering portfolio v14]

Source: Nordic BLE Engineering Portfolio v14, pages 3-9; firmware testing on separate Nordic development hardware.

CONNECTED SENSING AND WI-FI

ESP32-S3 · Connected Sensing & Wi-Fi Telemetry



Custom 41 x 41 mm PCB layout in KiCad, including GNSS, sensing and expansion.

```
void sendJson(int code, const String &json) {
  server.sendHeader("Access-Control-Allow-Origin", "*");
  server.sendHeader("Access-Control-Allow-Methods", "GET, OPTIONS");
  server.sendHeader("Access-Control-Allow-Headers", "Content-Type");
  server.send(code, "application/json", json);
}

String statusJson() {
  const bool connected = WiFi.status() == WL_CONNECTED;
  String json = "{\"device\":\"SenseLink X ESP32-S3 Rev A\",\"sensor_setup\":\"development\",\"uptime_ms\
  json += String(millis());
  json += ",\"wifi_configured\":";
  json += credentialsConfigured ? "true" : "false";
  json += ",\"wifi_connected\":";
  json += connected ? "true" : "false";
  json += ",\"wifi_rssi_dbm\":";
  json += connected ? String(WiFi.RSSI()) : String("null");
  json += ",\"ip_address\":";
  json += connected ? String("\"") + WiFi.localIP().toString() + "\"" : String("null");
  json += ",\"last_sample_uptime_ms\":" + String(currentState->uptimeMs);
  json += "}";
  return json;
}

void connectWifi() {
  lastAttemptMs = millis();
  WiFi.begin(SENSELINK_WIFI_SSID, SENSELINK_WIFI_PASSWORD);
  Serial.println("Wi-Fi connection attempt started.");
}
}
```

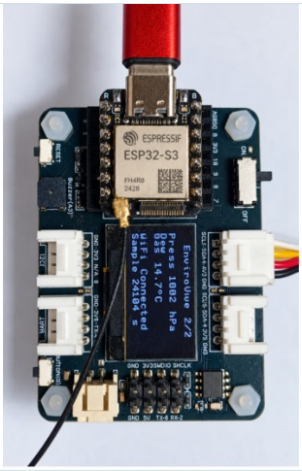
Wi-Fi connection handling and structured status/HTTP response implementation.

Hardware integration

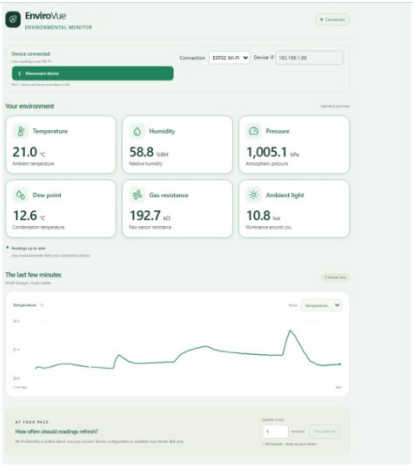
MAX-M10S GNSS, BMI323 IMU, ADXL372 high-g sensing, BME688 and VEML6035 interfaces, battery power and accessible peripheral expansion.

Embedded C++ and connectivity

PlatformIO/Arduino application with separate communications, display, sensor-state and environmental-maths modules. JSON carries sensor values, device status and heartbeat.



Physical ESP32-S3 development hardware with sensors and OLED output.



EnviroVue Wi-Fi client showing environmental values and recent trends.

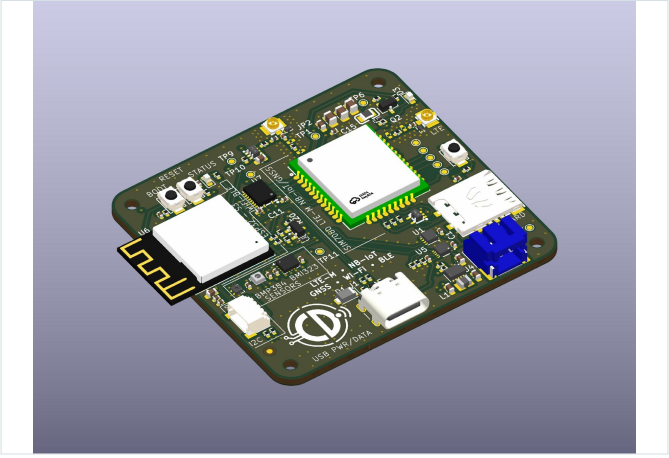
Evidence scope: firmware and application demonstrations used development hardware with BME680 and BH1750 sensors. They are separate from custom-PCB assembly testing. Wi-Fi is demonstrated.

[ESP32-S3 hardware and embedded engineering portfolio]

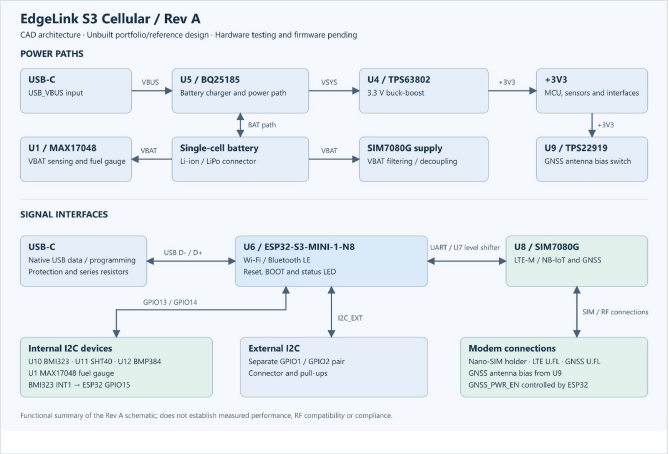
Source: ESP32-S3 Engineering Portfolio, pages 3-9; design, manufacturing preparation and development-platform testing.

CELLULAR SYSTEMS AND POWER ARCHITECTURE

EdgeLink S3 · Cellular & Battery-Powered Hardware



Rev A KiCad board render. CAD design, not a fabricated prototype.



Power and interface architecture linking the MCU, modem, sensors and battery system.

Charging, system power and modem supply

USB-C feeds a BQ25185 charger and power-path circuit. TPS63802 generates the 3.3 V logic rail; the modem uses a filtered battery supply. MAX17048 supports fuel gauging, with switched GNSS antenna bias.

Digital interfaces and wireless integration

Native ESP32-S3 USB, a level-shifted modem UART, Nano-SIM and LTE/GNSS antenna interfaces. Internal I2C connects BMI323, SHT40, BMP384 and the fuel gauge; external I2C remains separately accessible.

Placement and manufacturing documentation

Ground and supply layers support subsystem placement and routing. Editable KiCad sources sit alongside four-layer Gerbers, drill data, assembly BOM, CPL and readable architecture/schematic documents.

Documented handoff	Evidence
Assembly BOM	46 component groups with LCSC sourcing references
Placement data	89 fitted references, all on the top side
Fabrication and source	Four-layer Gerbers, drill files and editable KiCad project

Current status: Rev A CAD and manufacturing preparation completed.

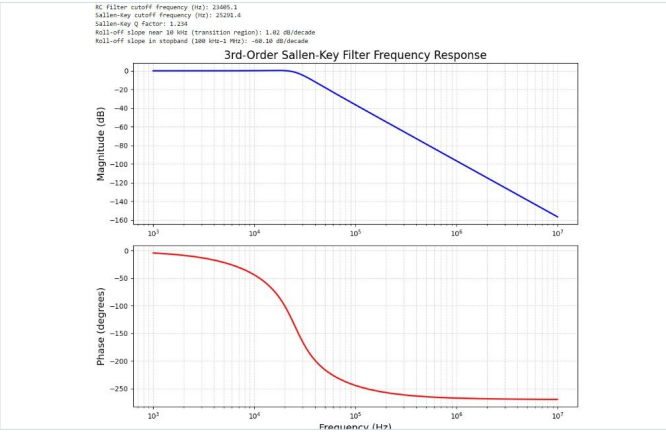
[EdgeLink S3 Cellular engineering portfolio]

Source: EdgeLink S3 Cellular Engineering Portfolio, pages 1-6; manufacturing counts as recorded in the report.

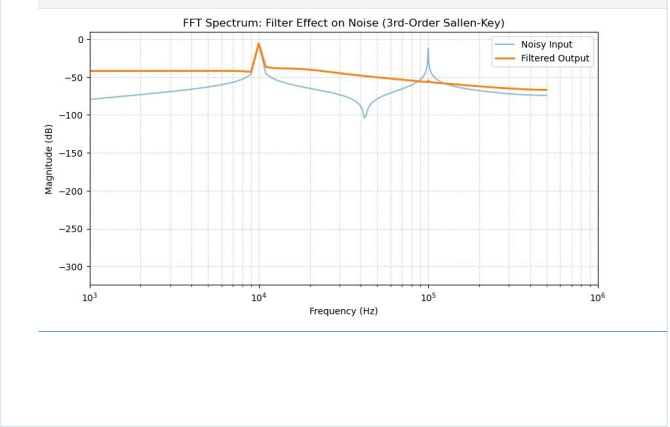
SIMULATION AND ENGINEERING ANALYSIS

Engineering Analysis, Simulation & Verification

LTspice and Python signal analysis support analog and RF design decisions, with supporting engineering



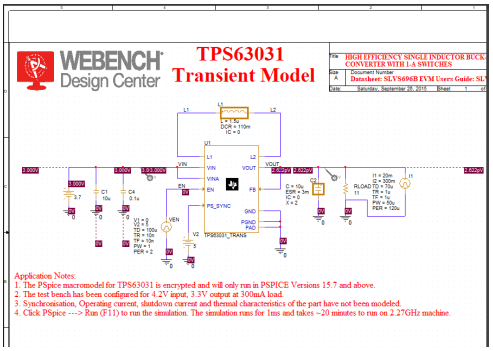
STM32WL Python magnitude/phase analysis of the analog low-pass network.



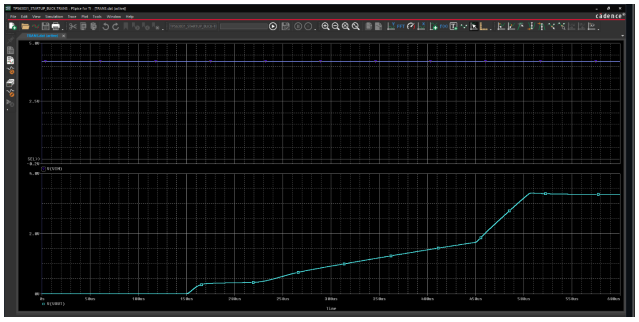
FFT test: wanted 10 kHz component retained; injected 100 kHz content attenuated.

Analog and RF simulation

LTspice AC and transient checks were compared with Python/NumPy/SciPy results. RF models explored the 868 MHz match and the trade-off between output filtering, passband loss and second-harmonic rejection.



TPS63031 PSpice transient model: startup and power-stage test circuit.



PSpice transient response: simulated VOUT startup and settling behaviour.

Model-based analysis; waveform is not a physical bench measurement.

Sources: STM32WLE5 Rev 2.0, sections 6-10 and figures 11/13/22; ESP32-S3 portfolio page 7.

SYSTEM INTEGRATION / CONNECTED PRODUCT ENGINEERING

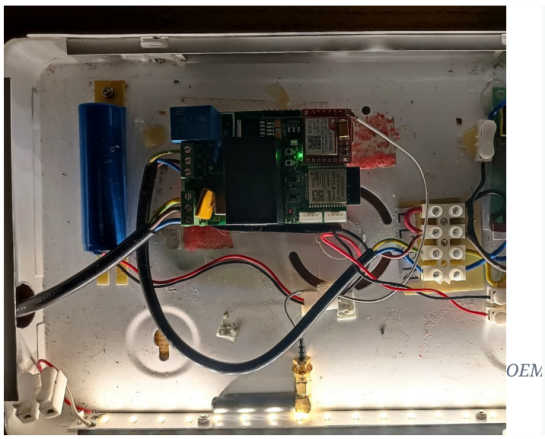
ExitLight | Automated Emergency-Light Testing & Cellular IoT Integration

ESP32-based retrofit integrating relay control, battery-backed light testing, BH1750 sensing, SIM800L GPRS and AWS IoT/MQTT reporting.

PHYSICAL PROTOTYPE & OEM RETROFIT

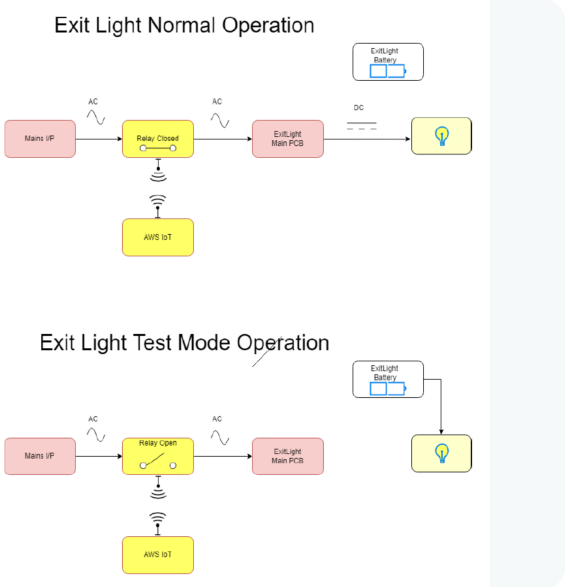


Working controller PCB

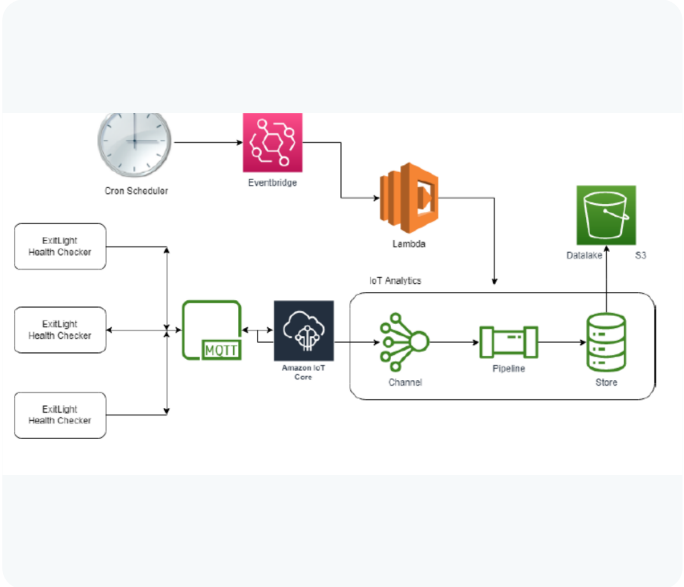


Controller installed within an existing emergency-light fitting

SYSTEM BEHAVIOUR & REMOTE MONITORING



Normal operation / battery-backed test mode



EventBridge, Lambda, AWS IoT Core and data flow

ENGINEERING SCOPE

Hardware and mains relay integration | Embedded ESP32 control | I2C light sensing
Cellular telemetry | Cloud scheduling and records | Physical retrofit validation

Evidence: working controller, OEM integration, operating-mode and AWS diagrams

PRODUCTION ENGINEERING / NPI & MANUFACTURING HANDOVER

From Design to Manufacturing Readiness

Production engineering, NPI and controlled manufacturing information packs.

CONTROLLED REV A HANDOVER · LAN7800 EXAMPLE

The LAN7800 USB 3.0 / Gigabit Ethernet adapter provides an inspectable example of how an Altium design is prepared for supplier review, purchasing and PCB assembly.

01

Engineering baseline

Schematic, PCB source, stack-up, DRC and reviewed design records.

02

Fabrication data

Gerber X2, drill data, fabrication drawings and PCB requirements.

03

Component control

BOM, AVL / sourcing references and component traceability.

04

Assembly handover

Assembly drawings, component placement and CPL data.

05

Test preparation

Inspection / test planning and blank records; not measured results.

06

Document control

Revision register, package traceability and approval status.

CROSS-FUNCTIONAL RELEASE PROCESS



ENGINEERING EVIDENCE & RELEASE BOUNDARY

Commercial experience includes manufacturing support, trial and pre-series builds, production documentation and ISO 9001 cross-functional coordination (FAAC Electronics).

The LAN7800 Rev A portfolio package demonstrates documented manufacturing preparation. Its document-control register is Rev 1.0.

Project evidence: github.com/CDonohoe-Designs/USB3-Gigabit-Ethernet-Adapter-Portfolio

PRODUCTION SUPPORT / QUALITY & ENGINEERING CHANGE

DFM, DFT & Production Problem Solving

Engineering decisions that improve assembly robustness, test access and field reliability.

DESIGN FOR MANUFACTURE & TEST · PRACTICAL CONTROLS

01

DFM / Assembly

Footprints, polarity/orientation, placement constraints, fabrication checks and supplier queries.

02

DFT / Test access

Rail measurement access, programming interfaces, inspection points and test-procedure readiness.

COMMERCIAL FAILURE INVESTIGATION · FAAC ELECTRONICS

RF performance degradation

Investigation traced reduced RF performance to an inferior substituted inductor. Component identity and purchasing control were key to the corrective response.

PCB fabrication plating change

Dendritic-growth shorts followed a plating-process change. Investigation supported a production halt and supplier corrective action, connecting failure analysis with quality control.

ENGINEERING CHANGE & FEEDBACK LOOP

Detect > **Isolate** > **Root cause** > **Correct** > **Verify**

These are documented commercial engineering investigations.

SENIOR ELECTRONICS ENGINEERING / PRODUCT DEVELOPMENT

Engineering Leadership & End-to-End Product Delivery

Technical ownership from system requirements and electronic design through verification, product introduction and manufacturing support.

01

TECHNICAL LEADERSHIP

Technical decisions, design reviews, requirements, risk assessment and cross-functional coordination.

02

HARDWARE DEVELOPMENT

System architecture, analog and RF electronics, power, high-speed PCB design and embedded integration.

03

VERIFICATION & PROBLEM SOLVING

Circuit analysis, bench investigation, failure diagnosis, root-cause analysis and corrective action.

04

PRODUCT INTRODUCTION

Design transfer, supplier coordination, production documentation, test readiness and quality support.

ENGINEERING RESPONSIBILITY ACROSS THE LIFECYCLE

Requirements | **Architecture** | **Development** | **Verification** | **NPI** | **Production**

ENGINEERING PROFILE

25+ years of commercial electronic product engineering, technical problem-solving, qualification and manufacturing support, complemented by current hands-on design work.

Selected evidence: FAAC volume NPI | Watt Footprint pilot | AmbiSense reliability

Current hardware portfolio: Altium, KiCad, RF, wireless and embedded systems

ENGINEERING PORTFOLIO

https://github.com/CDonohoe-Designs/PCB-Electronics-Engineering-Portfolio

LinkedIn: linkedin.com/in/caoilte-donohoe-69601a40a/

CROSS-PROJECT ENGINEERING EVIDENCE

Engineering Capability & Evidence Index

Each capability is tied to a concrete design, analysis or development-hardware example. Detailed reports are linked below.

Capability	Evidence to inspect
Altium Designer	STM32WL schematics, copper/planes and DRC; USB stack-up, impedance and routing-rule screenshots
Analog and mixed-signal design	STM32WL protected input, buffering, anti-alias filtering, precision reference and external ADC
RF and positioning interfaces	STM32WL matching/filtering and via fencing; GNSS antenna bias; cellular antenna and SIM interfaces
Power architecture	STM32WL analog/digital supplies; Nordic buck-boost; EdgeLink charger, power path and modem rail
Circuit and signal analysis	LTspice AC/transient and RF models; Python/Jupyter Bode, time-domain and FFT comparisons
Embedded C and Zephyr/NCS	Nordic modular firmware, sensor integration, logging, settings and custom BLE services
Embedded C++ and Wi-Fi	ESP32-S3 PlatformIO/Arduino firmware, JSON/HTTP interface and live telemetry
PCB release and handoff	Gerber/drill, BOM, CPL, placement review, source CAD and manufacturing documentation
Physical application integration	Nordic BLE and ESP32-S3 Wi-Fi sensor-to-client demonstrations on development hardware

Tool emphasis: Altium Designer for STM32WL and USB/Ethernet. KiCad for Nordic, ESP32-S3 and EdgeLink. Circuit analysis uses LTspice and Python/Jupyter; firmware uses Zephyr/NCS and PlatformIO.

Detailed project evidence

STM32WLE5 hardware design summary

USB3 to Gigabit Ethernet repository

Nordic BLE engineering portfolio v14

ESP32-S3 wireless and embedded portfolio

EdgeLink S3 Cellular hardware portfolio

Main engineering portfolio · GitHub

LinkedIn · Professional profile

Evidence compiled from supplied project reports and screenshots. Updated portfolio - 14 October 2026.